GGG-B-471b
September 3, 1964

SUPERSEDING
Fed. Spec. GGG-B-471a
December 4, 1953

FEDERAL SPECIFICATION

BLADE, COPING SAW; AND BLADE, JEWELERS' SAW

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers saw blades for straight and contour cutting of ferrous and nonferrous materials, plastic, wood, bone, gold, etc.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varities of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* Blades shall be of the following types and classes, as specified (see 6.1):

Type I.—Blades, saw, coping.
Class 1.—Pin end.
Class 2.—Loop end.
Type II.—Blades, saw, jewelers'.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification:

Federal Specification:

GGG-F-671—Frames, Saw, Hand (Hack, Jewelers', Coping).
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-585—Boxes; Wood Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 5110

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Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Steel.* Blades shall be of carbon steel having a minimum carbon content of 0.50 percent, and hardened to show a Rockwell C of 42 (minimum) for cutting wood, and 55 (minimum) for cutting metal (see 6.1).

3.2 Finish. Blades shall have a good commercial finish free from defects which may impair their serviceability. Blades shall be treated to retard corrosion.

3.3 Illustrations. The illustrations shown herein are for the convenience of identification, and are not intended to preclude the purchase of blades which are otherwise in accordance with this specification.

3.4 Type I, coping saw blades.

3.4.1 *Class 1 blades, pin end.* Class 1 coping saw blades shall be designed with a pin in each end so that the pins will engage and tension the blade into the coping saw and triangular hack-saw frames specified in GGG-F-671. Blades shall conform to dimensions shown in table I and shall be similar to figure 1.

TABLE I.—*Type I, classes 1 and 2 blades, coping*

Nominal length		Blade[1] width	Blade[2] thickness	Teeth[1]
Loop	Pin			
Inches	*Inches*	*Inch*	*Inch*	*Per Inch*
6	6	0.0625	0.020	18
6	6	.093	.020	16
6	6	.110	.020	10, 15
6	6	.125	.021	10, 15, 16, 20
6	6	.250	.022	15
6½	6½	.070	.020	15, 20, 32
6½	6½	.085	.020	15
6½	6½	.093	.020	16
6½	6½	.110	.020	10, 15
6½	6½	.125	.021	10, 14, 15, 20
6½	6½	.250	.022	16

[1]Plus or minus 5 percent allowable.
[2]Plus or minus 10 percent allowable

3.4.2 *Class 2 blades, loop end.* Class 2 coping saw blades shall be designed with a loop on each end of blade; the loops when engaged into the coping saw frame will tension the blade. Blades shall conform to dimensions shown in table I and be similar to figure 2.

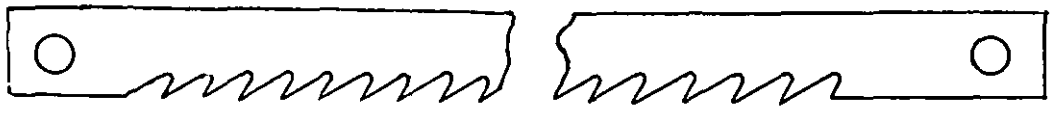


FIGURE 1.—Type I, class 1, pin end blades, coping.

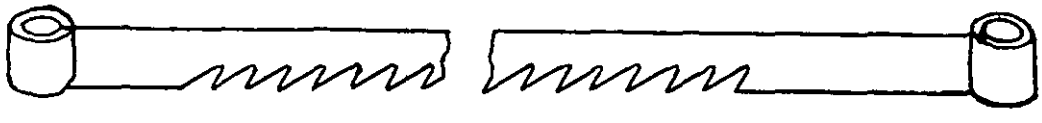


FIGURE 2.—Type I, class 2, loop end blades, coping.

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3.5 **Type II blades, saw, jewelers'.** Jewlers' saw blades shall be designed with flat blank ends for clamping into the jaws of the jewelers' saw frames specified in GGG-F-671. The blades shall conform to the dimensions shown in table II and shall be similar to figure 3.

TABLE II.—*Type II, blades, saw, jewelers'*

Size No.	Nominal length	Blade[1] width	Blade[2] thickness	Teeth[3]
	Inches	*Inch*	*Inch*	*Per inch*
8/0	5	0.012	0.006	96
7/0	5	.014	.007	80
6/0	5	.015	.008	76
5/0	5	.016	.008	78
4/0	5	.018	.009	70
3/0	5	.020	.010	66
2/0	5	.021	.010	64
1/0	5	.024	.011	64
1	5	.025	.012	56
2	5	.028	.014	48
3	5	.029	.014	44
4	5	.032	.015	40
5	5	.033	.015	36
6	5	.041	.019	32
7	5	.044	.022	30
8	5	.046	.023	20
10	5	.054	.016	30
12	5	.054	.016	20
14	5	.070	.020	15

[1]Plus or minus 5 percent allowable.
[2]Plus or minus 0.001 tolerance allowable.
[3]Plus or minus 5 percent allowable on 56 teeth or more; 8 percent allowable on 48 teeth or less.

3.6 **Workmanship.** Workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for quality conformance inspection.** Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.1 *Inspection lot.* All blades of the same part number shall be considered a lot for purposes of inspection.

4.2.2 A random sample of blades from each lot shall be examined at inspection level I, with an acceptable quality level (AQL) of 4.0 percent defective. Defects shall be classified in accordance with the defects set forth in table III. Any blade in the sample containing one or more defects shall be rejected, and if the number of defective blades in any sample lot exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE III.—*Classification of defects*

Examine	Defect
General	Indication of rust.
	Not corrosion-resistant treated.
	Not uniformly finished.
	Deep machine marks, burrs or pits.
Workmanship and contruction	Loose pins.
	Pins missing.
	Operation omitted or not properly performed.
	Blade bent, bowed or otherwise malformed.
	Cracked or fractured.
	Ends not properly shaped.

4.2.3 *Examination for dimensions and teeth count.* A random sample of blades shall be selected from each lot at inspection



FIGURE 3.—Type II, blades, saw jewelers'.

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level S-2 with an AQL of 2.5 percent defective to determine compliance with teeth count and dimensional requirements of tables I and II as applicable. Any deviation therefrom shall constitute a defect. If the number of defective blades in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 **Inspection of preparation for delivery requirements.** An inspection shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table IV. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE IV.—*Classification of preparation for delivery defects*

Examine	
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, Bulging or distortion of container.
Contents	Quantity per container not as specified.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and applications of the levels of preservation and packaging and packing shall be in accordance with Fed. Std. No. 102)

5.1 **Preservation and packaging.** Preservation and packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 *Level A.* The blades shall be preserved and packaged in accordance with MIL-H-15424, as specified for level A.

5.1.2 *Level B.* Twelve blades of one type and class shall be packaged in one of the following containers. Selection of the container shall be at the option of the contractor:

Specifications — *Title*

PPP-B-566—Boxes, Folding, Paperboard
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, Set-Up Paperboard.

Closure and sealing shall be in accordance with the applicable container specification or appendix thereto.

5.1.2.1 *Intermediate packaging.* Twelve packages of blades, packaged as specified in 5.1.2 shall be intermediate packaged in a box conforming to PPP-B-636, type I, class 1, style RSC. Closure and sealing shall be in accordance with the appendix to the box specification.

5.1.3 *Level C.* Preservation and packaging which will afford adequate protection against corrosion, deterioration, and damage during shipment from supply source to the first receiving activity for immediate use (vendor to user).

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Blades of one type and class shall be packed in accordance with MIL-H-15424, as specified for level A.

5.2.2 *Level B.* Twelve intermediate packages of blades, of one type and class packaged as specified in 5.1.2.1, shall be packed in one of the following containers. Selection of the container shall be at the option of the contractor.

Specification — *Type or Class*

PPP-B-585—Class 1 or 2 use
PPP-B-591—Domestic type
PPP-B-601—Domestic type
PPP-B-621—Class 1
PPP-B-636—Type I, class 1

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Closure and sealing shall be in accordance with the appendix to the applicable container specification.

5.2.3 *Level C.* Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to markings required by the contract or order, the unit and intermediate packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military activities.* In addition to markings required by the contract or order, the unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Class, length, width, thickness and number of teeth for type I blades; size number of type II blades (see 1.2 and tables I and II).

(*c*) Quantity.

(*d*) Metal-cutting blades or wood-cutting blades (see 3.1.1).

(*e*) Level of packaging and packing required (see 5.1 and 5.2).

(*f*) Special marking, if required (see 5.3).

6.2 Illustrations. The illustrations herein are descriptive and not restrictive, and are not intended to preclude the purchase of blades otherwise conforming to this specification.

6.3 Transportation description. Transporation description applicable to these items is:

Saw blades (saws without frames or handles) not otherwise indexed by name:
Carload minimum weight 30,000 pounds.
Truckload minimum weight 30,000 pounds.

MILITARY CUSTODIANS:

Army—GL

Navy—Ships

Air Force—MOA

Preparing Activity:

GSA-FSS

Copies of this specification may be purchased for 5 cents each.

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